

Modern Application Development-I

Project Report

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About Me : I am a BS in Data Science and Application student of IIT Madras, and currently I'm in the diploma level, and along with this degree I am also pursuing BS in Information Technology from offline college, and currently I am in the 3rd year.

Project Title : Placement Portal Application

Problem Statement :

To design and build a web-based application that allows Admin (Institute), Companies and Students to interact with the system based on their roles, for smooth campus recruitment.

Approach :

The app was built using Flask as the backend framework with a modular structure. I have thoroughly followed the project live sessions and taken the reference from there and built the placement portal application. The approach was, first the templates were created using the HTML, CSS and bootstrap, then 3 models (User, PlacementDrives, Application) was created and the relationships were setup, followed by the database creation and then routes was created to perform CRUD operations, etc.

AI/LLM Declaration :

I have used ChatGPT (Go) to assist in writing some routes, SQLAlchemy model definitions, and improve the variable naming consistency. The extent of AI/LLM usage is around 10-15% limited to code suggestions.

All final implementation logic, debugging and the integration were done manually.

Technologies and Frameworks Used :

<u>Technology / Library</u>	<u>Purpose</u>
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
SQLite	Lightweight local database for storing user data
Matplotlib	Generates charts (pie chart, bar chart, etc.)

Database Schema / ER Diagram

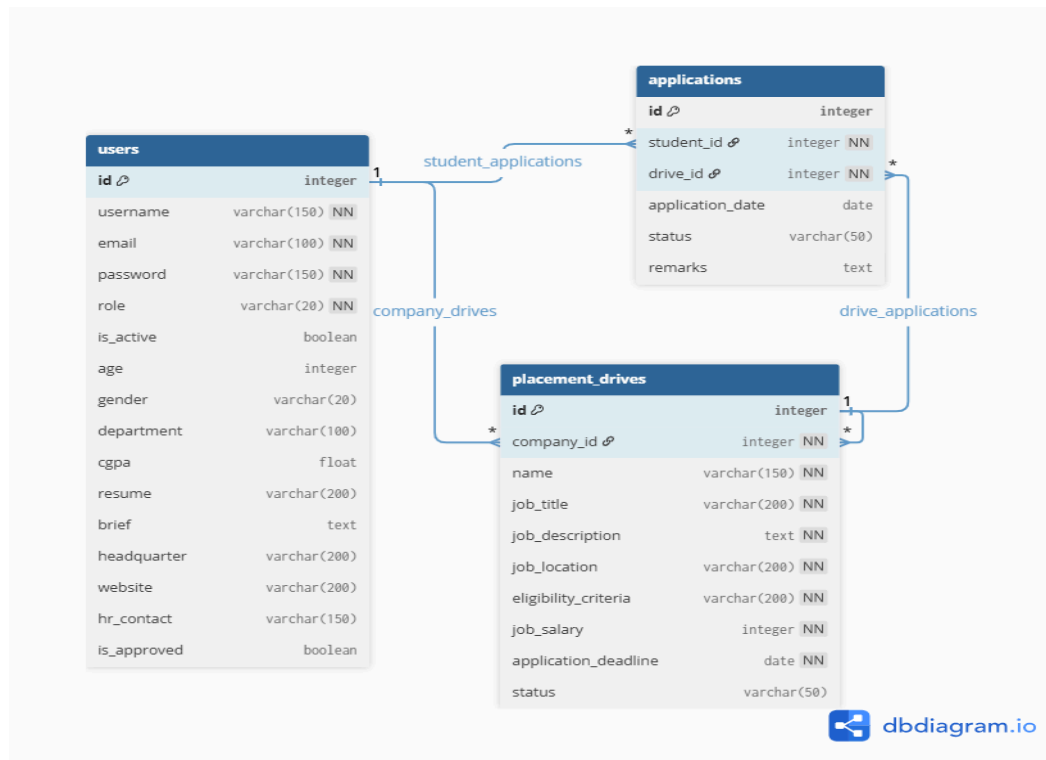
Tables :

1. **User** — Stores the profile details of Admin, Companies and Students (id, username, email, password, role, etc.)
2. **PlacementDrive** — Stores all the details of the placement drive (drive name, job title, job location, job salary, application deadline, status, etc.)
3. **Application** — Stores all the details of the Job application (student id, drive id, submission date, status of the application, remarks, etc.)

Relationships :

- **One-to-Many** → Company → Placement Drives
- **One-to-Many** → Student → Applications
- **One-to-Many** → Placement Drive → Applications
- **Many-to-Many** → Students ↔ Drives

ER Diagram :



Video Presentation :

Drive link : 📁 Video Report MAD-I Project.mp4